

Perturbation Theory

Preliminaries

Consider a system that we know how to solve. That is, consider a Hamiltonian H for which we know the eigenvalues and eigenstates *exactly*. In other words, we know what all quantities in the following equation are:

$$\hat{H} |n\rangle = E_n |n\rangle. \quad (1)$$

The label n suggests that what we really know is a collection of solutions to this eigenvalue problem: $\{E_n, |n\rangle\}$ for $n = 0, 1, 2, \dots$. That is, we know the full *spectrum* of H . While you may have seen several such examples in your coursework, such as the free particle, the particle in a box, the harmonic oscillator, and the hydrogen atom, these are all idealized systems that we do not expect to be faithful characterizations of the real world. So far, we have only been able to *hope* that they provide rough approximations to the real world. In this section of the course, we will be able to go beyond this hope and actually *quantify* how good these approximations are by modelling departures of our idealized systems from reality (or rather, from the more complex theory that we think ought to better describe reality) as small perturbations to our idealized Hamiltonian.

How do we know, for example, that our calculation for the Hydrogen atom is accurate enough to be useful? You may worry about the fact that our force was exactly Coulombic, that we neglected the spin of the electron, that we neglected relativistic effects, or the finite size of the nucleus, and so on. In all these cases, we do not expect the effect to completely invalidate our simplified model, but we should still be able to ask: *by how much does our solution, namely the energy levels and wavefunctions, of the Hydrogen atom change once we incorporate effect X or Y?* Our job in the following lectures will be to describe this procedure using *perturbation theory*. The idea is that our simplified calculation ought to be correct up to small “corrections” that do not modify our answers too much. If the corrections are large, they are no longer “corrections”, they become the dominant physics of our problem and we are better off solving the full problem from scratch. One may have to resort to numerical methods or to approximations that are not perturbative in nature, such as the variational method or the WKB approximation. However, if the corrections are small, then we can view the problem as a series expansion in *some* small parameter, to be defined by the problem at hand, which when sent to zero will recover our idealized solution.



Perturbation theory is an approximation method that allows us to find approximate solutions to problems that can recover an exactly-solvable problem in the limit that a small parameter goes to zero. For the most part, this method is an exercise in series expansions and in bookkeeping, but it is very powerful and a pillar to a variety of areas of physics, from the physics of Feynman diagrams to gravitational waves.



You can read the discussion in S&N Chapter 5. For now, we will focus on the time-independent case, which is the main focus of Sections 5.1 to 5.3. Shankar Chapter 17 also covers this material in very similar ways.

Time-Independent Perturbation Theory (Non-Degenerate Case)

Let us set up the problem more concretely. Imagine we start with the idealized Hamiltonian H_0 . I will now use the subscript 0 to emphasize the idealized nature of it. The spectrum of H_0 is known,

$$\hat{H}_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle. \quad (2)$$

Now, let us imagine that we can model a departure from this idealized system by adding a small correction to this Hamiltonian. What does small mean in this case? If my new Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{V}, \quad (3)$$

then I do not know what small means since $\hat{V} < \hat{H}$ does not make sense; $\hat{H}$ and $\hat{V}$ are operators. If it is numbers that we wish to compare, then we can say that all matrix elements of $\hat{V}$ in some basis are small compared to the corresponding matrix elements of $\hat{H}_0$ in that same basis. If that is true, then we say that $\hat{V}$ is a small perturbation to $\hat{H}_0$. To make this more precise, however, we can introduce a parameter λ that will always show up multiplying $\hat{V}$, so that whenever we calculate some expectation value of $\hat{V}$, we will have a factor of λ in front of it:

$$\hat{H} = \hat{H}_0 + \lambda \hat{V}. \quad (4)$$

Setting λ to zero recover our idealized system and setting it to one recovers the full solution that we are interested in. Mathematically, λ is a bookkeeping device that allows us to keep track of the order of our perturbation expansion.



In some examples, the form of $\hat{V}$ may already suggest what a good choice of λ is. For example, if we see a small parameter as a multiplicative constant in front of some other operator, such as $\hat{V} = \lambda \hat{W}$, then we can use this factor λ as our bookkeeping device. In this case, however, if we are to describe the real world, we will have to contend with whatever value of λ is given to us. Note that for our approximation method to be successful, it will not be enough to say that λ is small in this case because it is a constant we are not free to send to zero; what perturbation theory wants from us is for the expectation value of $\hat{V}$ to be small compared to the expectation value of $\hat{H}_0$ in some basis (see more below). For that to happen, we will have to check that the expectation value of $\hat{W}$ does not overcome the smallness of λ in our problem. It is easy enough, however, to just pretend λ is a free parameter in our calculations and then set it whatever value it takes (or to one if introduced by hand). At that point we can then check that the correction we calculation is indeed small.

Let us start then with the eigenvalue problem for $\hat{H}$ starting from the one for $\hat{H}_0$. The question is how do we find the eigenvalues and eigenstates of $\hat{H}$ given that we know the eigenvalues and eigenstates of $\hat{H}_0$?

$$\hat{H}_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle \xrightarrow{?} \hat{H} |n\rangle = E_n |n\rangle, \quad \text{where} \quad \hat{H} = \hat{H}_0 + \lambda \hat{V}. \quad (5)$$

We will think of the spectrum of $\hat{H}$ as a deformation of the spectrum of $\hat{H}_0$, and a deformation that is a function of the real parameter λ . It makes sense then to write the eigenvalues and eigenstates of $\hat{H}$ as

$$|n(\lambda)\rangle \quad \text{and} \quad E_n(\lambda) \quad (6)$$

to explicitly indicate the λ dependence of the spectrum of $\hat{H}$. Written in this way, we can make contact with our first step in the approximation: expand the eigenvalues and eigenstates of $\hat{H}$ as a power series in λ .

$$|n(\lambda)\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots \quad (7)$$

$$E_n(\lambda) = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \quad (8)$$

The coefficients of these power series are what we call the *perturbative corrections* to the eigenvalues and eigenstates of $\hat{H}_0$. The first order correction to the eigenstate $|n^{(0)}\rangle$ is $|n^{(1)}\rangle$, the second order correction is $|n^{(2)}\rangle$, and so on. The first order correction to the eigenvalue $E_n^{(0)}$ is

$E_n^{(1)}$, the second order correction is $E_n^{(2)}$, and so on. Because the energies are just real numbers, we can genuinely think of the perturbation expansion for the energies as a Taylor expansion in λ around $\lambda = 0$:

$$E_n(\lambda) = E_n^{(0)} + \lambda \left. \frac{\partial E_n}{\partial \lambda} \right|_{\lambda=0} + \frac{\lambda^2}{2!} \left. \frac{\partial^2 E_n}{\partial \lambda^2} \right|_{\lambda=0} + \dots \quad (9)$$

The perturbation expansion for the eigenstates is a bit more subtle because the eigenstates are vectors in a Hilbert space, but it is similar in spirit. This form is not very helpful because we do not E_n to begin with.

We can now plug our expansion of Eqs. (7) and (8) into the eigenvalue equation for $\hat{H}$. Formally, we have $\hat{H} |n(\lambda)\rangle = E_n(\lambda) |n(\lambda)\rangle$, which we write explicitly as

$$(\hat{H}_0 + \lambda \hat{V}) (|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \dots) = (E_n^{(0)} + \lambda E_n^{(1)} + \dots) (|n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \dots). \quad (10)$$

When $\lambda \rightarrow 0$, we recover the eigenvalue equation for $\hat{H}_0$ as expected. Treating λ as a free expansion parameter, it must be the case that this equality holds order by order in λ (if that does not make sense to you, just imagine the case where $\lambda = 10^{-100}$).

Order by order in λ , we have different equations

$$\lambda^0 : \hat{H}_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle, \quad (11)$$

$$\lambda^1 : \hat{H}_0 |n^{(1)}\rangle + \hat{V} |n^{(0)}\rangle = E_n^{(0)} |n^{(1)}\rangle + E_n^{(1)} |n^{(0)}\rangle, \quad (12)$$

$$\lambda^2 : \hat{H}_0 |n^{(2)}\rangle + \hat{V} |n^{(1)}\rangle = E_n^{(0)} |n^{(2)}\rangle + E_n^{(1)} |n^{(1)}\rangle + E_n^{(2)} |n^{(0)}\rangle, \quad (13)$$

and so on. We do not learn anything new with at order λ^0 , but at order λ^1 we can rearrange the equation to get

$$(\hat{H}_0 - E_n^{(0)}) |n^{(1)}\rangle = (E_n^{(1)} - \hat{V}) |n^{(0)}\rangle. \quad (14)$$

Note how the mismatch between 0-subscripted and 1-subscripted quantities in this equation. In particular, note how we do not know what $\hat{H}_0 |n^{(1)}\rangle$ is nor what $\hat{V} |n^{(0)}\rangle$ is, so this form is not particularly useful. At this point, we then take the inner product of both sides with $\langle n^{(0)}|$ to get

$$\langle n^{(0)}| (\hat{H}_0 - E_n^{(0)}) |n^{(1)}\rangle = \langle n^{(0)}| (E_n^{(1)} - \hat{V}) |n^{(0)}\rangle \implies \boxed{E_n^{(1)} = \langle n^{(0)}| \hat{V} |n^{(0)}\rangle}. \quad (15)$$

where we used the hermiticity of the Hamiltonian on the left-hand side to get $\langle n^{(0)}| \hat{H}_0 = E_n^{(0)} \langle n^{(0)}|$ and used $\langle n^{(0)}|n^{(0)}\rangle = 1$ on the right-hand side. This is the first order correction to the energy eigenvalue $E_n^{(0)}$. It suffices to know the unperturbed eigenstate $|n^{(0)}\rangle$ and the perturbation $\hat{V}$ to calculate this correction. Our full solution to the problem is then

$$E_n = E_n^{(0)} + \langle n^{(0)}| \hat{V} |n^{(0)}\rangle + \mathcal{O}(\lambda^2), \quad (16)$$

where we have set $\lambda = 1$ since it is just a bookkeeping device.

How about the first order correction to the eigenstate $|n^{(1)}\rangle$? To find this, we first write our first-order state in terms of the zeroth-order ones:

$$|n^{(1)}\rangle = \sum_m c_m |m^{(0)}\rangle, \quad (17)$$

where we have introduced the coefficients c_m to be determined. Note that I can do this even if the spectrum of $\hat{H}_0$ is degenerate. As long as the spectrum of $\hat{H}_0$ is complete, we can write any state in the Hilbert space as a linear combination of the eigenstates of $\hat{H}_0$. The degenerate

case just means that such a decomposition is not unique. Now, we can plug this decomposition into the first-order equation to get

$$\sum_m c_m (\hat{H}_0 - E_n^{(0)}) |m^{(0)}\rangle = (E_n^{(1)} - \hat{V}) |n^{(0)}\rangle, \quad (18)$$

which we can take the inner product of both sides with $\langle m^{(0)}|$ to get

$$c_m (E_m^{(0)} - E_n^{(0)}) = \langle m^{(0)}| (E_n^{(1)} - \hat{V}) |n^{(0)}\rangle. \quad (19)$$

For $m = n$, this equation is satisfied by the first-order correction to the energy that we already found, so it does not give us any information about c_n . For $m \neq n$, we can solve for c_m ,

$$c_m = \frac{\langle m^{(0)}| \hat{V} |n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}}, \quad (20)$$

which finally gives us the first-order correction to the eigenstate $|n^{(1)}\rangle$,

$$|n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle m^{(0)}| \hat{V} |n^{(0)}\rangle}{E_n^{(0)} - E_m^{(0)}} |m^{(0)}\rangle. \quad (21)$$

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What happened to the $m = n$ term in the sum? The $m = n$ term is not fixed by the first-order equation, so we are free to choose it as we please. The standard choice is to set $c_n = 0$, which is equivalent to choosing $|n^{(1)}\rangle$ to be orthogonal to $|n^{(0)}\rangle$. This is a convenient choice because it allows us to keep the normalization of our state fixed order by order in perturbation theory. To see this, calculate the norm of $|n(\lambda)\rangle$:

$$\langle n(\lambda)|n(\lambda)\rangle = \langle n^{(0)}|n^{(0)}\rangle + 2\lambda \Re(\langle n^{(0)}|n^{(1)}\rangle) + \lambda^2 \langle n^{(1)}|n^{(1)}\rangle + \dots \quad (22)$$

which can be easily satisfied at $\mathcal{O}(\lambda)$ by choosing $|n^{(1)}\rangle$ to be orthogonal to $|n^{(0)}\rangle$ since we started out with $\langle n^{(0)}|n^{(0)}\rangle = 1$.

Inspecting the expression above, we can also suggest an alternative method to normalize the state $|n(\lambda)\rangle$ at $\mathcal{O}(\lambda)$, namely by choosing $\langle n^{(0)}|n^{(1)}\rangle$ to be purely imaginary, which would also make the $\mathcal{O}(\lambda)$ term in the norm vanish. This means that the correction in the direction of the n -th unperturbed eigenstate is purely a phase correction. You can see that by going back to Eq. (7), but using a new state $|n^{(1)}(\lambda)'\rangle$

$$|n^{(1)}(\lambda)'\rangle = e^{i\alpha\lambda} |n^{(1)}(\lambda)\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + i\alpha\lambda |n^{(0)}\rangle + \dots \quad (23)$$

where we expanded in λ accounting also for the arbitrary complex phase $e^{i\alpha\lambda}$. This rephasing of the state leaves the physics unchanged (all observables are invariant under this rephasing). Using Eq. (21) for $|n^{(1)}\rangle$, we can see that the $\mathcal{O}(\lambda)$ correction in the direction of $|n^{(0)}\rangle$ is $i\alpha\lambda |n^{(0)}\rangle$, which is purely imaginary as we wanted. In conclusion: we can also normalize our states order by order by choosing the correction in the direction of the unperturbed state to be purely imaginary. So, choosing c_n to be zero is actually equivalent to choosing the correction in the direction of the unperturbed state to be purely imaginary since these two scenarios are related by a rephasing of the state.

Note that there is a bit of a choice in this procedure. As it stands, we have chosen to normalize our states order by order in perturbation theory, but we could have chosen to normalize them differently. For example, we could have chosen to normalize our states at the end of the day, after summing all the corrections up to a given order, which would have given us a different expression for the first-order state.

At second order in λ , we can follow a similar procedure to find the second-order corrections to the energy and to the state. Using the short-hand notation $\langle m^{(0)} | \hat{V} | n^{(0)} \rangle = V_{mn}$, the final result is

$$E_n^{(2)} = \sum_{m \neq n} \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}, \quad (24)$$

$$|n^{(2)}\rangle = \sum_{m \neq n} \left[\sum_{k \neq n} \frac{V_{mk} V_{kn}}{(E_n^{(0)} - E_m^{(0)})(E_n^{(0)} - E_k^{(0)})} - \frac{V_{mn} V_{nn}}{(E_n^{(0)} - E_m^{(0)})^2} \right] |m^{(0)}\rangle. \quad (25)$$

Again, the absence of the $m = n$ term in the sum for $|n^{(2)}\rangle$ is due to our choice of normalization for the state $|n(\lambda)\rangle$.

One may worry about the many terms in the sums above. To calculate a correction to a given energy level we have to sum over all the other energy levels, which may be an infinite number of them. Fortunately, there are two reasons why this procedure is not as bad as it looks. First, for higher and higher energy levels, the denominator $E_n^{(0)} - E_m^{(0)}$ becomes larger and larger, so the contribution of these terms to the sum becomes smaller and smaller (in particle physics, we call this *decoupling*). Second, in many cases the perturbation $\hat{V}$ is such that the matrix elements V_{mn} are only important for a few values of m and n , so this sum can be effectively truncated to a finite number of terms.

Note that for the ground state $n = 0$, the total correction to the energy up to second order is

$$E_0 = E_0^{(0)} + \langle 0^{(0)} | \hat{V} | 0^{(0)} \rangle + \sum_{m \neq 0} \frac{|V_{0m}|^2}{E_0^{(0)} - E_m^{(0)}} + \dots \quad (26)$$

From this expression we can deduce that 1) adding the first-order correction overestimates the true energy of the ground state, and 2) the second-order correction is always negative since $E_0^{(0)} - E_m^{(0)} < 0$ for all $m \neq 0$. The former result is a consequence of the variational principle.

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We can rewrite the correction to the ground state energy to first order as

$$E_0^{(0)} + \langle 0^{(0)} | \hat{V} | 0^{(0)} \rangle = \langle 0^{(0)} | \hat{H}_0 + \hat{V} | 0^{(0)} \rangle = \langle 0^{(0)} | \hat{H} | 0^{(0)} \rangle \geq E_0, \quad (27)$$

so the first-order correction to the ground state energy is an upper bound on the true ground state energy as guaranteed by the variational principle when we interpret $|0^{(0)}\rangle$ as a trial state for the full Hamiltonian $\hat{H}$.

An important point about these expressions is that the perturbation is mixing unperturbed states together through the matrix elements of $\hat{V}$. This *mixing* is precisely why degeneracies are a problem for perturbation theory: if $E_n^{(0)} = E_m^{(0)}$ for some $m \neq n$, then the mixing between the states $|n^{(0)}\rangle$ and $|m^{(0)}\rangle$ becomes very large, and the perturbation expansion breaks down. In other words, a term of our expansion blows up and we no longer have a small correction to our idealized solution. Fortunately, this blowing up problem can be easily fixed by working in a better basis, namely the one that diagonalizes $\hat{V}$ in the degenerate subspace of $\hat{H}_0$. This is what we will do next.

Time-Independent Perturbation Theory (Degenerate Case)

The degenerate case requires more care. Inspecting Eq. (21), we find that our only hope to avoid the blow-up problem is to have $\langle m^{(0)} | \hat{V} | n^{(0)} \rangle = 0$ for all m such that $E_m^{(0)} = E_n^{(0)}$. Note that the $m = n$ case is dealt with separately (i.e., through a choice of the normalization of the state, as discussed above), so we only have to worry about true degeneracies, which are the ones

that involve different states $m \neq n$. This will be exactly the way out: to apply perturbation theory to a degenerate system, we first have to find a basis of states that diagonalizes $\hat{V}$ in the degenerate subspace of $\hat{H}_0$. In other words: we must rotate our basis of states in the degenerate subspace of $\hat{H}_0$ to a new basis where

$$V_{mn} = \langle m^{(0)} | \hat{V} | n^{(0)} \rangle = 0 \quad \text{if} \quad E_m^{(0)} = E_n^{(0)} \text{ for } m \neq n. \quad (28)$$

To see how this comes about, let us start with the simplest example imaginable: a two-dimensional degenerate subspace of $\hat{H}_0$ spanned by the states $|1^{(0)}\rangle$ and $|2^{(0)}\rangle$ with the same energy $E^{(0)} = E_1^{(0)} = E_2^{(0)}$. In the basis of these two states, the unperturbed Hamiltonian is just:

$$H_0 = E^{(0)} \mathbb{1}. \quad (29)$$

Clearly, any linear combination of $|1^{(0)}\rangle$ and $|2^{(0)}\rangle$ is an eigenstate of $\hat{H}_0$ with the same eigenvalue $E^{(0)}$ since:

$$\hat{H}_0 |\psi\rangle = E^{(0)} |\psi\rangle \quad \text{for any} \quad |\psi\rangle = a |1^{(0)}\rangle + b |2^{(0)}\rangle. \quad (30)$$

Now, we add the perturbation $\hat{V}$ to this system in the same basis, which we can write as

$$\hat{H} = \hat{H}_0 + \lambda \hat{V} \longrightarrow H = E \mathbb{1} + \lambda \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix}, \quad (31)$$

We want to find the first-order correction to the spectrum of this system, which is fully degenerate. Suppose we blindly use:

$$E^{(1)} = \langle \psi | \hat{V} | \psi \rangle = a^* a V_{11} + a^* b V_{12} + b^* a V_{21} + b^* b V_{22}, \quad (32)$$

What should we pick for a and b ? That is, what is the correct choice of the state $|\psi\rangle$ in the degenerate subspace of $\hat{H}_0$ to calculate the first-order correction to the energy? And the energy of what?

To resolve this, let us recall that the degenerate states of the unperturbed system span the entire space of this particular problem. Therefore, the solutions for the *full* Hamiltonian are restricted to this degenerate subspace and we can write any state in the full problem as a linear combination of the degenerate states of the unperturbed problem:

$$|\psi\rangle = a |1^{(0)}\rangle + b |2^{(0)}\rangle \quad \Longleftrightarrow \quad \vec{c} \equiv \begin{pmatrix} a \\ b \end{pmatrix}. \quad (33)$$

The exact eigenvalue equation restricted to this subspace is

$$\hat{H} |\psi\rangle = E |\psi\rangle \quad \Longleftrightarrow \quad (E^{(0)} \mathbb{1} + \lambda V) \vec{c} = E \vec{c}, \quad (34)$$

where V is the 2×2 matrix with entries $V_{ij} = \langle i^{(0)} | \hat{V} | j^{(0)} \rangle$. Now expand the exact energy as in Eq. (8),

$$E = E^{(0)} + \lambda E^{(1)} + \lambda^2 E^{(2)} + \dots. \quad (35)$$

Plugging this into the eigenvalue equation and canceling the $E^{(0)} \mathbb{1}$ term on both sides gives

$$\lambda V \vec{c} = \lambda E^{(1)} \vec{c} + \mathcal{O}(\lambda^2). \quad (36)$$

Dividing by λ and keeping only the leading (first-order) terms yields the key result:

$$V \vec{c} = E^{(1)} \vec{c}. \quad (37)$$

In other words, the *first-order energy shifts* $E^{(1)}$ are precisely the eigenvalues of the perturbation matrix V restricted to the degenerate subspace, and the corresponding eigenvectors $\vec{c}$ determine

the *correct zeroth-order linear combinations* of the degenerate unperturbed states. This is why the ambiguity in $|\psi\rangle$ becomes a diagonalization problem: the unperturbed degeneracy means there is no preferred basis within the degenerate subspace, and the perturbation $\hat{V}$ selects the preferred basis by diagonalization.

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From the above we learn the following: if the perturbation $\hat{V}$ is itself diagonal in the degenerate subspace of $\hat{H}_0$, then we can just use the unperturbed states to calculate the first-order correction to the energy, and we do not have to worry about the blow-up problem. The simplest case is if $\hat{V}$ is proportional to the identity in the degenerate subspace, in which case we continue to have a degeneracy in the spectrum of $\hat{H}$ and we can just use the unperturbed states to calculate the first-order correction to the energy.

If $\hat{V}$ is not diagonal, finding its eigenbasis tells us how to smoothly deform the non-degenerate spectrum of $\hat{H}$ into the degenerate spectrum of $\hat{H}_0$ as we send $\lambda \rightarrow 0$.

In this extremely simple example, we can easily solve the eigenvalue problem in Eq. (37) and find the difference between the two first-order energy shifts $E_1^{(1)}$ and $E_2^{(1)}$ to be

$$E_1^{(1)} - E_2^{(1)} = \sqrt{(V_{11} - V_{22})^2 + 4|V_{12}|^2}. \quad (38)$$

This is the famous *level splitting* phenomenon: the degeneracy of the unperturbed system is lifted by the perturbation, and the energy levels split apart by an amount that depends on the matrix elements of the perturbation in the degenerate subspace. A diagonal perturbation ($V_{12} = 0$) will split the levels according to $|V_{11} - V_{22}|$, while an off-diagonal perturbation ($V_{11} = V_{22}$) will split the levels according to $2|V_{12}|$.

The generalization of this procedure to higher-dimensional degenerate subspaces is straightforward: we just have to diagonalize the perturbation $\hat{V}$ within the degenerate subspace of $\hat{H}_0$ to find the correct linear combinations of the degenerate states and the corresponding first-order energy shifts.

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What about if our spectrum contains non-degenerate and degenerate levels at the same time? In that case, we can apply the non-degenerate perturbation theory to the non-degenerate levels and the degenerate perturbation theory to the degenerate levels separately. Indeed, diagonalizing $\hat{V}$ in the degenerate subspace does not affect the non-degenerate levels, and the non-degenerate levels do not affect the degenerate levels at first order in perturbation theory since

$$E_n^{(1)} = \langle n^{(0)} | \hat{V} | n^{(0)} \rangle \quad (39)$$

for a non-degenerate level n . Iterating this procedure order by order in perturbation theory, we can find the corrections to all the energy levels and eigenstates of $\hat{H}$, both degenerate and non-degenerate, as long as we are able to diagonalize $\hat{V}$ in the degenerate subspaces of $\hat{H}_0$.



Convergence: Now we can address one subtlety about the convergence of our perturbation expansion. Even if we find that the expectation value of $\hat{V}$ is small compared to the expectation value of $\hat{H}_0$ in some basis, we may be tempted to celebrate the success of our perturbation expansion, but that would be premature. To see what can go wrong, let us consider the case where $\hat{V}$ is diagonal in the basis of eigenstates of $\hat{H}_0$ and has a small expectation value compared to $\hat{H}_0$ in that basis. In this case, the first-order correction to the energy is small, but the second-order correction can be large if there are states m such that $E_n^{(0)} - E_m^{(0)}$ is small, even if V_{mn} is small. This is a common problem in perturbation theory: even if the perturbation is small, the corrections can be large if there are near-degeneracies in the spectrum of $\hat{H}_0$. In these cases, it may still be possible to use perturbation theory, but we can do so by treating the near-degenerate states as a degenerate subspace and diagonalizing $\hat{V}$ within that subspace, as we did for the truly degenerate case. This is only true in some cases, however, and in general the presence of near-degeneracies can signal a breakdown of perturbation theory, which may require a non-perturbative treatment to resolve.

Symmetry breaking in the Hydrogen atom.

The level splitting phenomenon is a manifestation of the fact that degeneracies are often a consequence of symmetries of the system, and perturbations that break these symmetries can lift the degeneracies. For example, consider the hydrogen atom, which has several symmetries (rotational and parity symmetries for example) that lead to degeneracies in the energy levels. When we add perturbations that break any of these symmetries, such as external B or E field, the spin-orbit coupling or the Lamb shift, we find that the degeneracies are lifted and the energy levels split apart. This is a common theme in physics: symmetries lead to degeneracies, and degeneracy lifting indicates symmetry breaking perturbations.

Zeeman effect

Imagine that we have a hydrogen atom in the absence of any perturbation. Ignoring fine structure, the unperturbed energies depend only on n , so there is a large degeneracy within each fixed n level, including both the n degeneracies in $\ell = 0, \dots, n-1$ and the $(2\ell+1)$ degeneracy in $-\ell < m < \ell$ for each ℓ . In this section, we will break each one of these degeneracies by breaking the symmetry that gave rise to them. The first is the most straightforward: rotational symmetry leads to the m degeneracy. The second, which explains why energy levels do not depend on ℓ , is a bit more subtle (see, e.g., S. Weinberg, “*The SO(4) Symmetry of the Hydrogen Atom*”), but can also be broken by departing from a pure Coulomb potential.

Now apply a uniform external magnetic field, which we take to point along the z -axis, $\vec{B} = B \hat{z}$. The Zeeman perturbation is

$$\hat{V}_Z = -\vec{\mu} \cdot \vec{B} \quad (40)$$

where the electron magnetic moment has an orbital and a spin piece,

$$\vec{\mu} = -\frac{e}{2m_e} (\vec{L} + g_s \vec{S}), \quad \text{with} \quad g_s \simeq 2, \quad (41)$$

(using the convention that the electron charge is $-e$ with $e > 0$). Therefore

$$\hat{V}_Z = \frac{e}{2m_e} (\vec{L} + g_s \vec{S}) \cdot \vec{B} = \mu_B B \left(\frac{\hat{L}_z}{\hbar} + g_s \frac{\hat{S}_z}{\hbar} \right), \quad (42)$$

where $\mu_B \equiv e\hbar/(2m_e)$ is the Bohr magneton. For now, we will only consider the external B field, and ignore the effective magnetic field generated by the electron’s motion around the nucleus (the spin-orbit coupling), which is a separate perturbation that we will discuss later.

(Normal) Zeeman effect: We begin with the simpler case where the degeneracy in the hydrogen spectrum is lifted by the orbital contribution alone. This case will illustrate how our perturbation theory for degenerate states works in practice when our unperturbed states are already diagonal in the perturbation. If we ignore spin (or consider a spinless particle), the B field perturbation reduces to

$$\hat{V}_Z = \mu_B B \frac{\hat{L}_z}{\hbar}. \quad (43)$$

In the basis $|n, l, m\rangle$, we have $\hat{L}_z |n, l, m\rangle = \hbar m |n, l, m\rangle$, so the perturbation is already diagonal:

$$\langle n, l, m | \hat{V}_Z | n, l, m' \rangle = \mu_B B m \delta_{mm'}. \quad (44)$$

Thus, the first-order Zeeman shift is

$$E_{n,l,m}^{(1)} = \mu_B B m. \quad (45)$$

Each fixed (n, l) level splits into $2l + 1$ equally spaced sub-levels labeled by $m = -l, -l + 1, \dots, l$ when an external B field is applied, and the spacing between adjacent sub-levels is $\mu_B B$.